

No. of Functional Features Included in the Solution

Date	1 July 2026
Team ID	
Project Name	Human Development Index (HDI) Prediction – A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	Accepts Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita from the user	Frontend (HTML)	Done	High
2	Input Validation	Validates user-entered values before prediction.	Flask Backend	Done	Medium
3	HDI Prediction	Uses the trained Linear Regression model to predict the HDI value.	Machine Learning Model	Done	High
4	HDI Category Classification	Classifies the predicted HDI into Low, Medium, High, or Very High Human Development.	Flask Backend	Done	High
5	Result Display	Displays the predicted HDI value, category, and progress bar.	Result Page (HTML/CSS)	Done	High
6	Model Loading	Loads the saved HDI.pkl model using Pickle.	Backend	Done	Medium
7	Responsive User Interface	Provides a clean and user-friendly interface using HTML and CSS.	Frontend	Done	Medium
8	Predict Again Feature	Allows users to perform multiple predictions without restarting the application.	Frontend	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8

Core / Must-Have Features	5
Additional / Nice-to-Have Features	3
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Input Form, Result Page, Predict Again Button
2	Backend / Logic	3	Input Validation, HDI Prediction, Category Classification
3	Database / Storage	1	Saved ML Model (HDI.pkl)
4	API / Integration	0	Not Used
5	Security / Authentication	0	Not Implemented